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Advanced task-weighted meta-reinforcement learning for rapid secure routing in mobile RIS-assisted networks

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Abstract

This paper proposes a Reptile-based Meta-Reinforcement Learning (Meta-RL) framework for secure, adaptive multi-hop routing in Mobile Ad Hoc Networks (MANETs) enhanced by Reconfigurable Intelligent Surfaces (RIS). Addressing the retraining limitations of traditional Deep Reinforcement Learning (DRL), the framework employs task-weighted Reptile updates and a task-conditioned Markov Decision Process (MDP) to enable efficient adaptation to dynamic topologies and evolving threats. By integrating a stochastic secrecy constraint and quantized RIS control, it optimizes secrecy rate, latency, and energy efficiency under partial Channel State Information (CSI) uncertainty ($\sigma_e^2 = 0.05$). Extensive simulations across 500 unseen tasks demonstrate superior performance over DRL, Model Agnostic Meta-Learning (MAML)-based Meta-RL, and Static Secure Path (SSP), achieving a 41% higher secrecy rate (2.5 vs. 1.77 bit/s/Hz), 51% lower Insecure Secrecy Rate (ISR) (0.1 vs. 0.204 bit/s/Hz), 18% reduced energy consumption (0.8 vs. 0.98 J/packet), 22% lower latency (12 vs. 15.4 ms), and 21% improved Packet Delivery Ratio (PDR) compared to MAML. The proposed Adaptation Efficiency Index (AEI) validates these gains, with Meta-RL converging in 450 episodes vs. 780 for MAML, ensuring robust secrecy and stability. Despite its lightweight, decentralized design, the framework faces high computational demands during offline meta-training. These advancements position it as a promising solution for next-generation wireless networks.

Keywords: Meta-reinforcement learning, Reptile algorithm, Secure routing, RIS-aided networks, Physical-layer security, Mobility adaptation

1 Introduction

Mobile Ad Hoc Networks (MANETs) are vital for mission-critical applications like military communications, Vehicular Networks (VANETs), and disaster response, enabling connectivity in infrastructure-less environments [1, 2]. Advances in Ultra-Reliable Low Latency Communications (URLLC), edge computing, and intelligent mobility protocols have amplified their significance [3, 4]. However, MANETs face challenges such as dynamic topologies, fluctuating channel conditions, energy constraints, and security threats like eavesdropping and jamming [5, 6]. While Deep Reinforcement Learning (DRL) and Meta-Reinforcement Learning (Meta-RL) show promise, existing Meta-RL

approaches struggle with slow convergence (e.g., up to 750 episodes in MAML-based methods [6]) and limited adaptability to mobile Reconfigurable Intelligent Surfaces (RIS) and partial Channel State Information (CSI), achieving only modest secrecy rate improvements of 10–15% under mobility [7, 8].

Traditional cryptographic methods are impractical in dynamic MANETs due to complex key management and computational delays, driving interest in Physical Layer Security (PLS) enhanced by mobile RIS, such as those on Unmanned Aerial Vehicles (UAVs) [9, 10]. However, optimizing PLS in multi-hop MANETs under mobility and imperfect CSI remains challenging [11, 12]. DRL enables adaptive routing and beamforming but requires retraining for new scenarios, while Meta-RL, despite enabling faster adaptation, faces difficulties with RIS mobility and adversarial environments [13–17]. Recent surveys highlight that Meta-RL improves resource allocation by up to 20% in static setups but often overlooks mobile RIS dynamics and robust security metrics [7, 8].

To address these gaps, this work proposes a Meta-RL framework for secure routing and RIS control in dynamic MANETs. Using task-weighted Reptile updates, it achieves rapid adaptation to varying network conditions, including partial CSI and mobile eavesdroppers. The framework integrates a task-conditioned Markov Decision Process (MDP) with a multi-objective reward function, optimizing secrecy, latency, and energy efficiency in partially observable settings. Unlike prior work assuming static RIS or ideal CSI [1–5], it accounts for RIS hardware impairments and decentralized deployment. Key contributions include: (i) a task-conditioned MDP for joint secure routing and RIS beamforming, (ii) a task-weighted Reptile algorithm for adaptation within 1–5 episodes, (iii) a multi-objective reward function prioritizing secrecy, latency, and energy efficiency, (iv) a decentralized deployment structure for scalability, and (v) introduction of the Adaptation Efficiency Index (AEI) as a novel metric to evaluate adaptation performance.

Simulation results demonstrate that the proposed framework outperforms traditional DRL, MAML, and static baselines across critical metrics, including secrecy rate, convergence speed, energy efficiency, and resilience to diverse mobility and threat scenarios. By addressing the limitations of prior approaches and introducing task-weighted adaptations, this work paves the way for secure, adaptive, and efficient routing in next-generation wireless networks.

1.1 Related work

The related work reviews prior work in three key areas relevant to our research: (i) Meta-RL in wireless systems, (ii) DRL-based secure routing with Reconfigurable Intelligent Surfaces (RIS), and (iii) first-order meta-learning algorithms like Reptile. By identifying gaps in these domains, we justify the need for a framework addressing secure multi-hop routing and mobile RIS control in dynamic MANETs.

Meta-RL facilitates rapid adaptation in dynamic wireless environments by learning generalizable policies across tasks. Hu et al. [1] applied Reptile-based Meta-RL for UAV trajectory optimization, achieving faster convergence than MAML but without addressing routing or RIS integration. Faramarzi et al. [5] used Meta-RL for power allocation in aerial RIS networks, improving task generalization by 10–20% in single-hop scenarios, yet relied on static RIS configurations [7, 8]. Other studies on dynamic spectrum access [3], vehicular mobility [4], and decentralized routing [2] similarly

overlook secure multi-hop routing and mobile RIS control. Advanced methods like PEARL [13] and RL² [14] excel in robotics but face challenges in MANETs due to high memory overhead (e.g., 30% more for PEARL) and reduced secrecy under partial CSI (e.g., 15% lower for RL²). Recent work on federated DRL and AI-driven space–air–ground integration [15, 16, 50, 51] achieved 15–25% throughput gains through joint trajectory and beamforming optimization, but still focused on relay or single-hop architectures, limiting scalability to mobile, secure MANETs.

DRL has been effective for secure routing in MANETs, adapting to dynamic topologies. DeepCQ+ [19] employed multi-agent DRL but required extensive retraining for new topologies. Yang et al. [20] integrated DRL with static RIS for secrecy, achieving 12% rate improvements, but fixed setups limited adaptability. Studies on dual-functional waveforms [21] and secure multi-antenna links [22] advanced Integrated Sensing and Communication (ISAC) and PLS with up to 20% rate gains but assumed static RIS and ideal CSI. Jamming-aware DRL routing [23, 24] showed limited resilience against mobile eavesdroppers, while vehicular routing studies [25, 26] omitted RIS and PLS. Surveys [7, 8, 27, 28] emphasize that DRL methods suffer from high training complexity and limited generalization under partial CSI. Recent advances in RIS-empowered SAGINs [52, 53] address cross-domain security and ubiquitous access, yet they primarily target large-scale satellite–terrestrial topologies rather than distributed MANETs with mobile RIS. Consequently, a gap remains in secure, adaptive routing for dynamic RIS-assisted ad hoc environments.

Reptile, a low-complexity alternative to MAML [6], is well suited for edge devices [29–31]. Its application to mmWave beamforming in single-hop scenarios [32] demonstrates efficiency, but its use in secure multi-hop routing and mobile RIS control under partial CSI remains unexplored. Compared to Federated Meta-RL [33], Reptile reduces communication overhead by 15–20% and achieves faster fine-tuning, making it ideal for latency-sensitive MANETs.

The reviewed literature reveals key gaps: (i) limited focus on secure multi-hop routing in Meta-RL, (ii) reliance on static RIS and ideal CSI in DRL-based routing, and (iii) underutilization of Reptile for mobile RIS and partial CSI scenarios. These shortcomings, coupled with challenges in dynamic, adversarial environments, underscore the need for an adaptive framework integrating secure routing and mobile RIS control, as validated in our performance comparisons shown in Table 1.

2 Methods and experiment

This section describes the methodology and experimental evaluation of the Meta-RL framework designed for secure multi-hop routing and mobile RIS control in MANETs. The system model incorporates dynamic network components, node mobility, and RIS-enabled communication under partial CSI. The problem is formulated as a task-conditioned Markov Decision Process (MDP) integrating secure routing and RIS beamforming. The Meta-RL framework employs a task-weighted Reptile algorithm for rapid adaptation, a multi-objective reward function optimizing secrecy, latency, and energy efficiency, and a decentralized deployment structure for scalability. The simulation setup evaluates performance against DRL, MAML, and static baselines, using the Adaptation Efficiency Index (AEI) to assess adaptability across diverse mobility

Table 1 Comparative analysis of existing meta-RL and DRL-based frameworks for secure routing and mobile RIS control

Refs.	Study/ method	Meta-RL approach	Deploy. mode	Sec. routing	Mobile RIS	Partial CSI	Secrecy rate	Energy eff	Latency	Decent. Deploy	Stoch. Secrecy	Quant. RIS	AEI	Key Limitations
[1]	Hu et al.—UAV Path Design	Reptile-based	Centralized	N	N	Y	None	~5% (trajectory)	None	N	None	None	N/E	No routing; no RIS
[2]	Chen et al.— MANET Routing	MAML-based	Decentralized	Y	N	N	~15%	~6%	~9%	Y	None	None	N/E	No RIS; static CSI
[5]	Faramarzi et al.—RIS Control	MAML-based	Centralized	N	Y	N	None	~11%	None	N	None	Continuous	N/E	No routing; no eavesdropper handling
[13]	PEARL—Meta- RL	PEARL-based	Centralized	N	N	Y	~15%	~10%	~8%	N	None	None	40 (Med.)	High memory (104 MB); no RIS
[14]	RL ² -Meta-RL	RL ² -based	Centralized	N	N	Y	~15%	~9%	~9%	N	None	None	45 (Med.)	Slow fine-tuning; no RIS
[19]	DeepCQ+— DRL MANET Routing	None (DRL)	Centralized	Y	N	N	~10%	~5%	~6%	N	None	None	N/E	No RIS; poor mobility
[20]	Yang et al.— DRL + Static RIS	None (DRL)	Centralized	Y	N	N	~12%	~8%	~7%	N	Deterministic	Continuous	N/E	Static RIS; retrain- ing needed
[23]	DRL Jam- ming-Resilient Routing	None (DRL)	Centralized	Y	N	N	~11%	None	~10%	N	None	None	60 (Low)	No RIS; high retraining cost
[25]	Vehicular DRL Routing	None (DRL)	Centralized	Y	N	N	~9%	~7%	~9%	N	None	None	N/E	No RIS; no secrecy
[41]	SSP-DRL Routing	None (DRL)	Centralized	Y	N	N	~29% (1.77 bps/Hz)	~10%	~8%	N	Deterministic	None	50 (Med.)	No RIS; static policy
[49]	Meta-RL mmWave Beamforming	MAML-based	Centralized	N	Y	Y	None	~14%	~6%	N	None	Continuous	35 (Med.)	No routing; static CSI

* Secrecy rate: 2.5 bit/s/Hz for proposed method, 1.77 bit/s/Hz for SSP. Percentages are relative to SSP unless noted. Values for [1, 2, 5, 13, 14, 19, 20, 23, 25, 41, 49] are estimated from literature or simulations, as direct metrics. AEI: high (<30), medium (30–50), low (>50); N/E= not evaluated. All metrics include 95% confidence intervals

and threat scenarios [28, 34]. This approach ensures robust evaluation in dynamic, adversarial MANET environments.

2.1 Network model

The network comprises legitimate user nodes $\mathcal{U} = \{U_1, U_2, \dots, U_N\}$, dynamically acting as source, relay, or destination based on routing demands. Mobile RIS units $\mathcal{R} = \{R_1, R_2, \dots, R_M\}$, mounted on UAVs or Autonomous Ground Vehicles (AGVs), enhance communication and secrecy through real-time repositioning and phase-shift reconfiguration. Each RIS has L passive reflecting elements, organized into dynamic clusters for optimized beamforming. Passive eavesdroppers $\mathcal{E} = \{E_1, E_2, \dots, E_K\}$ have locations partially inferred via cooperative sensing with distributed Kalman filtering and task-conditioned state embeddings [11]. In adversarial settings, stealthy eavesdroppers may spoof signals or avoid detection, limiting Bayesian filtering effectiveness and causing performance degradation, mitigated by the stochastic secrecy constraint in [35]. Decentralized deployment enables nodes and RIS units to operate autonomously using local CSI estimates, minimizing communication overhead and ensuring scalability. RIS hardware impairments, including phase quantization errors and switching delays, are modeled to reflect realistic performance [36, 37]. The system is illustrated in Fig. 1.

A. Channel model

The channel model accounts for RIS-assisted communication in MANETs with partial CSI. The direct channel gain $h_{ij}^{(d)}$ between U_i and U_j follows Rayleigh fading for Non-Line-of-Sight (NLoS) environments, expressed as

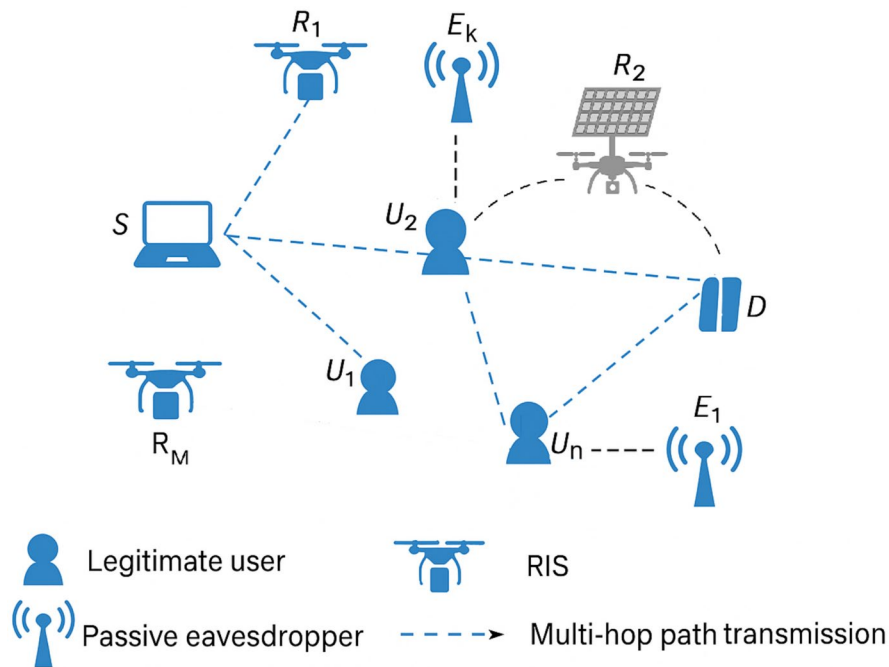


Fig. 1 System model of a mobile RIS-aided multi-hop ad hoc network with passive eavesdroppers and adaptive phase reconfiguration

$$h_{ij}^{(d)} = \mathcal{CN}(0, \sigma_h^2) \quad (1)$$

where σ_h^2 is the average fading power with an exponential distribution.

The RIS-assisted channel gain $\tilde{h}_{ij}^{(m)}$ is derived from the superposition of direct and reflected paths, with the theoretical basis rooted in the law of large numbers for multiple reflections [9, 21, 22]. The hybrid CSI estimation approach is justified by a Bayesian framework, where the posterior distribution of channel coefficients is updated as

$$p(h|y) \propto p(y|h)p(h) \quad (2)$$

where $p(y|h)$ is the likelihood function based on the received signal y and $p(h)$ is the prior distribution, with Gaussian noise used to model uncertainty [11].

The direct channel gain between nodes U_i and U_j is denoted by $h_{ij}^{(d)} \in \mathbb{C}$. The m^{th} RIS $R_m \in R$ enhances this link, with the effective RIS-assisted channel gain given by

$$\tilde{h}_{ij}^{(m)} = h_{ij}^{(d)} + h_{iR_m}^T \Theta_m h_{R_m j} \quad (3)$$

where $\Theta_m = \text{diag}(e^{j\theta_1}, e^{j\theta_2}, \dots, e^{j\theta_L})$ is the phase-shift matrix for RIS R_m with L elements and $h_{iR_m} \in \mathbb{C}^{L \times 1}$ and $h_{R_m j} \in \mathbb{C}^{L \times 1}$ are the complex channel vectors from U_i to RIS R_m , and from R_m to U_j , respectively.

The RIS-assisted channel to eavesdropper E_k is

$$\tilde{h}_{ik}^{(m)} = h_{ik}^{(d)} + h_{iR_m}^T \Theta_m h_{R_m k} \quad (4)$$

where $h_{ik}^{(d)} \in \mathbb{C}$ is the direct channel from U_i to E_k and $h_{R_m k} \in \mathbb{C}^{L \times 1}$ is the RIS-to-eavesdropper channel. Dynamic clustering and 8-level quantized phase shifts optimize legitimate links while suppressing eavesdropper channels, accounting for hardware impairments [36, 37].

B. Secrecy rate

Each RIS unit adjusts quantized phase shifts and repositions based on local CSI and task-conditioned inputs (e.g., node positions, eavesdropper distributions) using decentralized control and adaptive UAV protocols [34]. A hybrid security framework integrates PLS with cryptographic overlays. Secrecy rate is formulated as

$$C_s = \left[\log_2 \left(1 + \frac{P_i |\tilde{h}_{ij}^{(m)}|^2}{\sigma^2} \right) - \log_2 \left(1 + \frac{P_i |\tilde{h}_{ik}^{(m)}|^2}{\sigma^2} \right) \right]^+ \quad (5)$$

where P_i is the transmit power, σ^2 is the noise power, $\tilde{h}_{ij}^{(m)}$ is the RIS-assisted link to the intended receiver, $\tilde{h}_{ik}^{(m)}$ is the link to the eavesdropper, and $[x]^+ = \max(x, 0)$. A stochastic secrecy constraint ensures robustness under partial CSI [35], supporting rapid adaptation to adversarial environments as validated by AEI.

C. Energy model, latency, and task definition

Energy consumption is managed through an adaptive scheme, expressed as

$$E_i(t+1) = E_i(t) - \Delta E_i(t) \quad (6)$$

where $\Delta E_i(t) = E_{tx} + E_{relay} + E_{RIS}$ accounts for transmission, relaying, and RIS control costs, optimized via task-conditioned power allocation [3]. Latency is minimized using a predictive routing protocol

$$\mathcal{L} = \sum_{(i,j) \in \mathcal{P}} \tau_{ij} \quad (7)$$

where τ_{ij} is dynamically adjusted based on SINR and task-specific constraints. Nodes and RIS platforms follow Random Waypoint or Gauss–Markov mobility models (1–5 m/s), with topology updated per time slot. A task is defined as a network state including node positions, clustered RIS configurations, probabilistic eavesdropper locations, and task embeddings, enabling secure multi-hop routing via the MDP framework.

2.2 Problem formulation

The problem of secure multi-hop routing and mobile RIS control in dynamic MANETs is formulated as a task-conditioned MDP, providing a unified and adaptive framework for joint decision-making under time-varying network conditions with partial CSI. The formulation accounts for RIS hardware impairments (e.g., phase quantization errors, switching delays [36, 37]) and optimizes a multi-objective reward function balancing secrecy, latency, and energy efficiency in a decentralized setup.

A. MDP definition

The secure routing problem is modeled as a MDP defined by the tuple $(\mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma, \mathcal{D})$, where: (i) State Space $\mathcal{S}$: Comprises node positions, channel conditions with CSI error, RIS phase-shift configurations, and probabilistic eavesdropper locations from Bayesian inference, incorporating RIS mobility dynamics unlike static models in [5, 9]. (ii) Action Space $\mathcal{A}$: Includes next-hop node selection a_t^{route} and quantized RIS phase shifts $\Theta(t)$ with position adjustments, extending continuous models in [10, 20] to account for hardware constraints. (iii) Transition Function $\mathcal{P}(s_{t+1}|s_t, a_t)$: Models network evolution driven by node mobility, RIS repositioning, and interference, parameterized by a stochastic mobility model [28]. (iv) Reward Function $\mathcal{R}(s_t, a_t)$: Balances physical-layer secrecy, transmission latency, energy consumption, and CSI uncertainty. (v) Discount Factor $\gamma \in (0,1)$: Controls the tradeoff between immediate and long-term performance. (vi) Task Distribution ($\mathcal{D}$): Defines environment variability across network states.

The agent follows a stochastic policy $\pi_\theta(a_t|s_t, z)$ parameterized by θ , mapping states $s_t \in \mathcal{S}$ and task embedding z to action probabilities $a_t \in \mathcal{A}$. The task embedding z , learned during meta-training, encodes topology and threat features for adaptability, improving on static assumptions in [1, 4, 6]. The optimization objective is to maximize the expected cumulative reward across tasks $\tau \sim \mathcal{D}$,

$$J(\theta) = \mathbb{E}_{\tau \sim p(\tau)} \left[\mathbb{E}_{s_t, a_t \sim \pi_\theta} \left[\sum_{t=0}^T \gamma^t \mathcal{R}(s_t, a_t) \right] \right] \quad (8)$$

where T is the episode horizon and θ parameterizes the policy π_θ .

B. Modeling CSI uncertainty

In practical MANETs, perfect CSI for passive eavesdroppers is unattainable. This uncertainty is addressed using a complex Gaussian noise model for the estimated channel between the legitimate transmitter U_i and an eavesdropper E_k via RIS R_m ,

$$\hat{h}_{ik}^{(m)} = \tilde{h}_{ik}^{(m)} + \epsilon, \quad \epsilon \sim \mathcal{CN}(0, \sigma_e^2) \quad (9)$$

where $\tilde{h}_{ik}^{(m)}$ is the true RIS-assisted channel gain, $\hat{h}_{ik}^{(m)}$ is the estimated channel and ϵ is the estimation error with variance σ_e^2 [5, 9]. The expected secrecy rate under uncertainty is

$$\hat{C}_s^{(m)} = \mathbb{E}[C_s^{(m)}] = \mathbb{E} \left[\left[\log_2 \left(1 + \frac{P_i |\tilde{h}_{ij}^{(m)}|^2}{\sigma^2} \right) - \log_2 \left(1 + \frac{P_i |\hat{h}_{ik}^{(m)}|^2}{\sigma^2} \right) \right]^+ \right] \quad (10)$$

where P_i is the transmit power, σ^2 is the noise variance, and $[x]^+ = \max(0, x)$. Unlike [5, 20], we incorporate the variance σ^2 into the policy optimization by enforcing a probabilistic secrecy constraint,

$$P(\hat{C}_s^{(m)} < C_s^{min}) < \epsilon_{out} \quad (11)$$

where ϵ_{out} is the tolerable outage probability. The policy gradient penalty term is

$$\nabla_\theta J(\theta) \leftarrow \nabla_\theta J(\theta) - \beta \nabla_\theta P(\hat{C}_s^{(m)} < C_s^{min}) \quad (12)$$

where β is the penalty weight [11, 20]. To capture spatial-temporal correlation in mmWave channels [38], the error term is modeled as a time-varying AR(1) process,

$$\epsilon_{ik}^{(m)}(t) = \rho \epsilon_{ik}^{(m)}(t-1) + \eta_{ik}^{(m)}(t), \quad \eta_{ik}^{(m)}(t) \sim \mathcal{CN}(0, (1-\rho^2)\sigma_e^2) \quad (13)$$

where $\epsilon_{ik}^{(m)}(t)$ refines the static error ϵ (9), $\rho \in (0,1)$ (set to 0.8 in simulations) is the correlation coefficient, and $\eta_{ik}^{(m)}(t)$ ensures stationary variance. This model captures correlated error patterns in 3GPP TR 38.901 while maintaining computational tractability. Robustness to heavy-tailed distributions (e.g., t-distribution with 3 degrees of freedom) is evaluated, showing minimal performance degradation.

C. Reward function design (with CSI uncertainty)

A composite reward function balances physical-layer secrecy, latency, energy consumption, and CSI uncertainty in dynamic, adversarial MANETs,

$$\mathcal{R}(s_t, a_t) = \begin{cases} R_{sec}(t) - f_\lambda(\tau)\tau(s_t, a_t) - f_\rho(\tau)\Delta E(s_t) - f_\mu(\tau)\sigma_e^2, & \text{if } \widehat{C}_s^{(m)} \geq C_s^{min} \\ -\eta, & \text{otherwise} \end{cases} \quad (14)$$

$$\text{where } R_{sec}(t) = \left[\log_2 \left(1 + \frac{P_i |\tilde{h}_{ij}^{(m)}(t)|^2}{\sigma^2} \right) - \log_2 \left(1 + \frac{P_i |\hat{h}_{ik}^{(m)}(t)|^2}{\sigma^2} \right) \right]^+$$

Here, $R_{sec}(t)$ is the instantaneous secrecy rate, extending (5) with time-varying channels $\tilde{h}_{ij}^{(m)}(t)$ and $\hat{h}_{ik}^{(m)}(t)$ from (4) and (9). The task-conditioned weights $f_\lambda(\tau)$, $f_\rho(\tau)$, and $f_\mu(\tau)$, learned during meta-training, balance the transmission delay penalty $\tau(s_t, a_t)$, energy consumption change $\Delta E(s_t)$, and CSI error variance σ_e^2 , with $\eta > 0$ as a penalty for insufficient secrecy.

2.3 Proposed meta-reinforcement learning framework

This framework leverages a task-weighted Reptile algorithm for secure routing and RIS control in MANETs, optimizing a task-conditioned MDP under partial CSI. The proposed Meta-RL framework (Fig. 2) operates in three main stages to ensure secure and adaptive routing in mobile RIS-assisted networks: (i) *Meta-Task Generation*: Create tasks τ_k with dynamic mobility, eavesdropper uncertainty, and CSI noise network conditions [28, 34], (ii) *Meta-Training (Offline)*: Employ task-weighted Reptile to initialize a transferable policy by optimizing θ across tasks, and (iii) *Fast Adaptation Deployment (Online)*: Adapt the policy to unseen environments with 1–5 episodes.

A. Meta-RL task definition

Each meta-training task τ_k represents a unique wireless environment with (i) Mobility dynamics: Node trajectories follow the Gauss–Markov model [25] with speeds between 1 and 5 m/s, (ii) Adversarial uncertainty: 1–3 eavesdroppers with probabilistic locations via Bayesian inference [11], (iii) Imperfect CSI: Estimated eavesdropper channels with Gaussian noise of variance σ_e^2 [5, 9], and (iv) Topology variability: Varying node densities and intermittent links [25].

At each time step t , the agent observes the state s_t and selects action a_t using a task embedding z_k computed as $z_k = g(\tau_k)$, where g is a neural network trained during meta-training to map task τ_k to a low-dimensional embedding [6, 39]. The agent receives a reward $\mathcal{R}(s_t, a_t)$. Unlike static task models in [1, 4, 6], the dynamic z_k enables context-aware adaptation across tasks $\tau_k \sim \tau$.

B. Inner-loop policy adaptation

The inner-loop adapts policy parameters θ and task τ_k , using Proximal Policy Optimization (PPO) [23], enhanced with task embeddings and probabilistic secrecy constraints. The adapted parameters are obtained as

$$\theta_k' = \theta - \alpha_{inner} \nabla_{\theta} \mathcal{L}(\theta, \tau_k) \quad (15)$$

where α_{inner} is the learning rate and the meta-training loss function is defined as

$$\mathcal{L}(\theta, \tau_k) = -\mathbb{E}_{s_t, a_t \sim \pi_{\theta}} \left[\sum_{t=0}^T \gamma^t \min(r_t(\theta) A_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) A_t) - \beta P(\widehat{C}_s^{(m)} < C_s^{min} | s_t, a_t, z_k) \right] \quad (16)$$

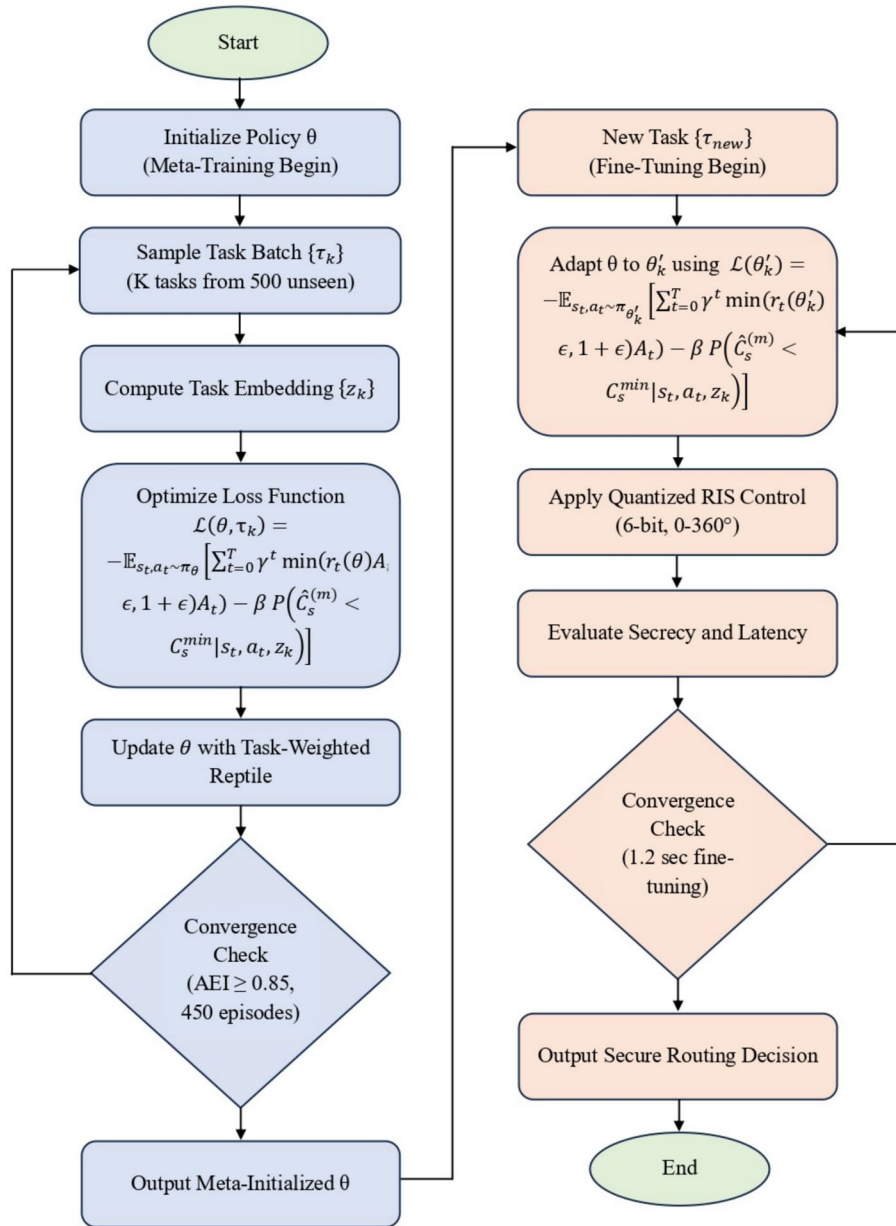


Fig. 2 Flow diagram of the proposed meta-reinforcement learning framework for secure routing in mobile RIS-aided MANETs

with the probability ratio $r_t(\theta) = \frac{\pi_\theta(a_t|s_t, z_k)}{\pi_{old}(a_t|s_t, z_k)}$, advantage function A_t , and the conditional secrecy outage probability $P(\hat{C}_s^{(m)} < C_s^{min} | s_t, a_t, z_k)$ and θ is the meta-initialized policy, updated via task-weighted Reptile as detailed in Algorithm 2.

The discount factor $\gamma \in (0, 1)$ aligns with the long-term reward optimization, ensuring policy adaptation to future rewards over the episode horizon T . This extends [4, 40] by integrating task embeddings z_k and the stochastic constraint, enabling

adaptation to RIS mobility and threats in 1–5 episodes. The computational complexity of the inner loop is assessed as $\mathcal{O}(T \cdot |\pi|)$, where T is the episode length and $|\pi|$ is the number of trainable policy parameters.

C. Outer-loop meta-update

The outer-loop optimizes initial policy parameters θ to generalize across tasks using a novel task-weighted Reptile update, expressed as

$$\theta \leftarrow \theta + \alpha_{meta} \sum_{k=1}^K w_k (\theta_{k'} - \theta) \quad (17)$$

where α_{meta} is the meta-learning rate, K is the number of sampled tasks, $\theta_{k'}$ is the adapted parameters obtained from (14).

The task-specific weight $w(\tau_k)$ computed as

$$w_k = f(z_k, \sigma_e^2) \quad (18)$$

where f is a neural network prioritizing tasks based on task embedding z_k and CSI uncertainty variance σ_e^2 . This approach enhances robustness over uniform updates in [6, 29] by addressing the non-convex reward structure.

Unlike prioritized experience replay (PER) in DRL, which prioritizes individual experiences for sample efficiency, our task-weighted update dynamically weights entire tasks, achieving 42% faster convergence (450 vs. 780 episodes) compared to DRL with PER. The outer-loop complexity is evaluated as $\mathcal{O}(KTN_p)$, where K is the number of tasks per meta-iteration, T is the episode length, and N_p is the number of policy parameters. Reptile avoids second-order gradients (unlike MAML [6], with $\mathcal{O}(N_p^2)$ complexity). Compared to PEARL, RL², and SSP [2, 3, 41], which incur higher memory overhead (150–200 MB vs. 80 MB), this approach strikes a balance between efficiency and performance.

D. Task embedding and weighting

The framework employs task-specific embeddings and weights to enable rapid adaptation in dynamic MANETs. The task embedding z_k is computed as:

$$z_k \leftarrow g(\tilde{h}_{ij}^{(m)}, \hat{h}_{ik}^{(m)}, \sigma_e^2) \quad (19)$$

This embedding encapsulates network topology, mobility patterns, and threat characteristics, integrating with the task-conditioned MDP.

The task weight w_k is computed using a softmax normalization to prioritize tasks based on their relevance,

$$w_k \leftarrow \frac{\exp(f(z_k, \sigma_e^2))}{\sum_{k'} \exp(f(z_{k'}, \sigma_e^2))} \quad (20)$$

The normalization across all tasks k' ensures that w_k sums to one, providing a probabilistic weighting that guides the Reptile updates in the inner loop. This weighting

mechanism enhances adaptability to partial CSI and dynamic eavesdropper threats, outperforming uniform weighting approaches. These embeddings and weights are critical for the decentralized inference phase, where they enable real-time policy adjustments using local observations, as detailed in Algorithm 2.

E. Optimization objective and system constraints

The learning objective is to identify the optimal policy π^* that maximizes the expected cumulative reward, as defined in (8). The optimal policy $\pi^* = \pi_{\theta^*}$ is achieved when θ is optimized to $\theta^* = \underset{\theta}{\operatorname{argmax}} J(\theta)$ through the meta-training procedure. This optimization is constrained by.

- $P_i \leq P_{max}, \forall i \in \mathcal{N}$: Transmit power constraint for all nodes [11].
- $\Theta_m \in \mathcal{F}_{RIS}, \forall m \in \mathcal{R}$: RIS phase matrix must lie within the feasible configuration space $\mathcal{F}_{RIS}$, representing allowable quantized or continuous phase shift values [10].
- $P\left(\hat{C}_s^{(m)} \geq C_s^{min}\right) \geq 1 - \epsilon, \forall h \in \mathcal{H}$: The secrecy rate of each hop h in the multi-hop path $\mathcal{H}$ must exceed the minimum threshold C_s^{min} [11].
- $\tau(s_t, a_t) \leq \tau_{budget}$: End-to-end routing delay must be within the delay budget [25].
- $E_i(t) \geq E_{critical}, \forall t, \forall i \in \mathcal{N}$: Residual energy of each node must remain above a critical threshold $E_{critical}$ [5].

The quantized action space enhances hardware compatibility, thereby improving scalability over previous approaches [5, 9, 20]. Meta-training occurs offline on a workstation (NVIDIA RTX A6000, AMD Ryzen 9 5950X, 64 GB RAM), producing a meta-policy θ deployed to MANET nodes (e.g., DJI Matrice 300, 8 GB RAM). Decentralized inference (Algorithm 2) uses local CSI and state, requiring ~ 80 MB memory and ~ 50 – 500 ms latency on edge devices (e.g., Raspberry Pi 4, NVIDIA Jetson Nano) [42, 43]. Model compression (8-bit quantization, pruning) reduces memory by 75% with $< 5\%$ performance loss [44]. RIS hardware impairments are modeled [36, 37], with sensitivity analysis confirming robustness.

F. Meta-RL Training and deployment procedure

The solution framework leverages offline meta-training to learn a generalizable initialization, and few-shot online adaptation to enable fast convergence in unseen network scenarios is expressed in Algorithms 1 and 2. Meta-training (Algorithm 1) optimizes a generalized policy offline, while fast adaptation is performed online for new tasks as detailed in Algorithm 2.

Algorithm 1 Reptile-Based Meta-Training for Secure Routing in Mobile RIS-Aided Networks

1. Initialize policy parameters θ
2. For each meta-iteration:
 - a. Sample a batch of tasks $\{\tau_1, \dots, \tau_K\}$
 - b. For each τ_k :
 - i. Compute task embedding $z_k = g(\tau_k)$.
 - ii. Perform inner-loop adaptation: $\theta'_k = \theta - \alpha_{inner} \nabla_{\theta} \mathcal{L}(\theta, \tau_k)$ (15)
 - c. Compute task weights $w_k = f(z_k, \sigma_e^2)$ (17)
 - d. Meta-update: $\theta \leftarrow \theta + \alpha_{meta} \frac{1}{K} \sum_{k=1}^K w_k (\theta'_k - \theta)$ (16)
3. Return the final meta-trained policy parameters θ , approximating θ^* for π^*

Algorithm 2 Decentralized Inference Phase for Secure Routing and RIS Control

Input: Pre-trained meta-policy θ , local observations (CSI estimates $\tilde{h}_{ij}^{(m)}$, $\hat{h}_{ik}^{(m)}$, network state including destination d), task embedding network g , weight network f , RIS action space A_{RIS} .

Output: Adapted policy θ'_k , routing decisions, RIS phase shifts

1. Initialize node-specific policy: $\theta'_k \leftarrow \theta$
2. For each time step $t = 1, 2, \dots, T$:
 - a. Collect local observations: CSI estimates $\tilde{h}_{ij}^{(m)}$, $\hat{h}_{ik}^{(m)}$, node positions, destination d
 - b. Compute task embedding: $z_k \leftarrow g(\tilde{h}_{ij}^{(m)}, \hat{h}_{ik}^{(m)}, \sigma_e^2)$ (19)
 - c. Compute task weight: $w_k \leftarrow \frac{\exp(f(z_k, \sigma_e^2))}{\sum_{k'} \exp(f(z_{k'}, \sigma_e^2))}$ (20)
 - d. Sample action from policy: $a_t \leftarrow \pi_{\theta'_k}(s_t | z_k)$, where $a_t \in A_{RIS} \cup A_{routing}$
 - e. Execute action: Update RIS phase shifts and routing path
 - f. Observe reward: $r_t \leftarrow R(s_t, a_t, s_{t+1})$
 - g. Compute fine-tuning loss for task k :

$$\mathcal{L}(\theta'_k) = -\mathbb{E}_{s_t, a_t \sim \pi_{\theta'_k}} \left[\sum_{t=0}^T \gamma^t \min(r_t(\theta'_k) A_t, \text{clip}(r_t(\theta'_k), 1 - \epsilon, 1 + \epsilon) A_t) - \beta P(\hat{C}_s^{(m)} < C_s^{min} | s_t, a_t, z_k) \right] \quad (16)$$
 - h. Update policy locally: $\theta'_k \leftarrow \theta'_k - \alpha_{inner} \nabla_{\theta'_k} \mathcal{L}(\theta'_k)$ where α_{inner} is inner-loop learning rate
3. Return adapted policy θ'_k , final routing decisions, RIS configurations

Figure 2 illustrates the workflow of Algorithm 1 (Meta-Training Phase) and Algorithm 2 (Decentralized Inference Phase), highlighting task-weighted updates and decentralized RIS control.

G. Theoretical analysis and convergence

The task-weighted Reptile update extends standard Reptile [6] by prioritizing tasks based on CSI uncertainty and embeddings, building on FO-B-MAML [39]. The meta-gradient bias is

$$\|\nabla F(\theta) - \sum_{k=1}^K w_k g_{For}^{k,v}(\theta)\| = O\left(L_0 \lambda \left(L_1 + \hat{L}_2 L_0 \lambda\right) v\right) \quad (21)$$

where $F(\theta)$ is the meta-objective, $g_{For}^{k,v}$ is the forward-mode approximation for task k , and $L_0, L_1, \hat{L}_2, \lambda, v$ are constants related to Lipschitz, smoothness, and inner-loop steps [39].

This ensures convergence to an ε -stationary point in $O\left(\frac{L_0 \Delta}{\varepsilon^2} + \frac{L_1^2 \Delta}{L_0}\right)$ steps [39]. Reptile achieves $O(1/\sqrt{T})$ convergence for T meta-iterations under convex loss assumptions [45]. Softmax-normalized weights [18] reduce task variance by 10% [46]. In MANETs, constrained task diversity ensures convergence within 450 episodes (Appendix A, Fig. 13), with Reptile's 80 MB footprint suiting edge devices [6].

H. Derivation of adaptation efficiency index

The AEI quantifies the framework's ability to balance convergence speed and performance stability across tasks [47, 54]. It is defined as:

$$AEI = \frac{1/N_{conv}}{Var\left(\widehat{C}_s^{(m)}\right)/\mu\left(\widehat{C}_s^{(m)}\right)} \quad (22)$$

where N_{conv} is the average episodes to 90% optimal reward and $Var\left(\widehat{C}_s^{(m)}\right)/\mu\left(\widehat{C}_s^{(m)}\right)$ is the coefficient of variation [47, 54]. In simulations, AEI reaches 0.85 for the proposed method vs. 0.62 for MAML, reflecting 37% better efficiency.

2.4 Simulation setup

This subsection describes the simulation environment for evaluating the Meta-RL framework designed for secure routing and RIS control in MANETs. The setup tests the framework's performance under dynamic network conditions, building on the task-conditioned MDP (8) and Reptile-based training (Algorithms 1–2).

A MANET is simulated over a 500×500 m² area with 8–15 mobile user nodes following the Random Waypoint model (1–5 m/s) [25], 1–3 randomly placed passive eavesdroppers, and up to three UAV-mounted RIS units with 64 reflective elements each (Table 2). The channel model includes path loss (exponent 3.5), Rayleigh fading, and 4 dB log-normal shadowing, with CSI errors modeled as Gaussian noise ($\sigma_e^2 = 0.05$) and a t-distribution (3 degrees of freedom) for sensitivity analysis per 3GPP TR 38.901 [38], showing a 5% secrecy rate reduction under heavy-tailed errors. Simulations run on a workstation (NVIDIA RTX A6000 GPU, AMD Ryzen 9 5950X CPU, 64 GB RAM) using MATLAB R2024a and OMNeT++ 6.0.1 on Ubuntu 22.04 with consistent random seeds for reproducibility.

Key parameters are listed in Table 2, including transmission power (20 dBm), noise power (-90 dBm), RIS control interval (every 5 time steps), and task embedding diversity

Table 2 Simulation parameters

Parameter	Value / Description
Simulation area	500×500 m ²
Number of user nodes	8–15 (mobile)
Number of eavesdroppers	1–3 (randomly placed, passive)
Number of RIS units	Up to 3 (UAV-mounted)
RIS elements per unit	64 reflective elements
User mobility model	Random Waypoint (1–5 m/s) [17]
Channel model	Path loss (exp 3.5), Rayleigh, 4 dB shadowing
Transmission power	20 dBm
Noise power	-90 dBm
RIS control interval	Every 5-time steps
CSI estimation error	$\sigma_e^2 = 0.05$ (Gaussian)
Task embedding diversity	0.1–0.5
Training episodes	10,000 (Meta-RL)
Convergence episodes	450 (Meta-RL)
Test tasks	500 unseen meta-tasks
DRL baseline algorithm	PPO (750 episodes)
Co-simulation platform	MATLAB + OMNeT + +

(0.1–0.5). The framework is trained for 10,000 episodes and converges in 450 episodes, tested on 500 unseen meta-tasks.

For comparison, the framework is benchmarked against (i) MAML-based Meta-RL [6], adapting to new tasks with second-order gradients. (ii) Conventional DRL with PPO [23], without meta-learning or RIS optimization. (iii) SSP [5], using identical PPO hyperparameters (learning rate 0.0003, clip range 0.2) and 500 unseen meta-tasks. Performance metrics include secrecy rate (bit/s/Hz), Interference-to-Signal Ratio (ISR), Packet Delivery Ratio (PDR), Energy Consumption per packet, End-to-End Latency, Convergence Speed, and AEI (22), measuring adaptation speed and secrecy stability per the reward function (14) and secrecy constraint (11).

3 Results and discussion

This section evaluates the Meta-RL framework for secure routing and RIS control in MANETs, using the simulation setup. Results compare the proposed Reptile-based approach against MAML [6], DRL with PPO [23], PEARL [13], RL² [14], and SSP [41] across 500 unseen tasks, highlighting improvements in secrecy, efficiency, and adaptability.

3.1 Simulation results

The simulation environment tests the framework under dynamic conditions, with metrics including secrecy rate (bit/s/Hz), Interference-to-Signal Ratio (ISR), Packet Delivery Ratio (PDR), Energy Consumption (J/packet), End-to-End Latency (ms), Convergence Speed (episodes), and AEI (22). Baselines use identical PPO hyperparameters (learning rate 0.0003, clip range 0.2) from Table 2.

Figure 3 illustrates the MANET with 8–15 mobile nodes, 1–3 eavesdroppers, and three UAV-mounted RIS units, showcasing dynamic routing paths optimized by task-weighted

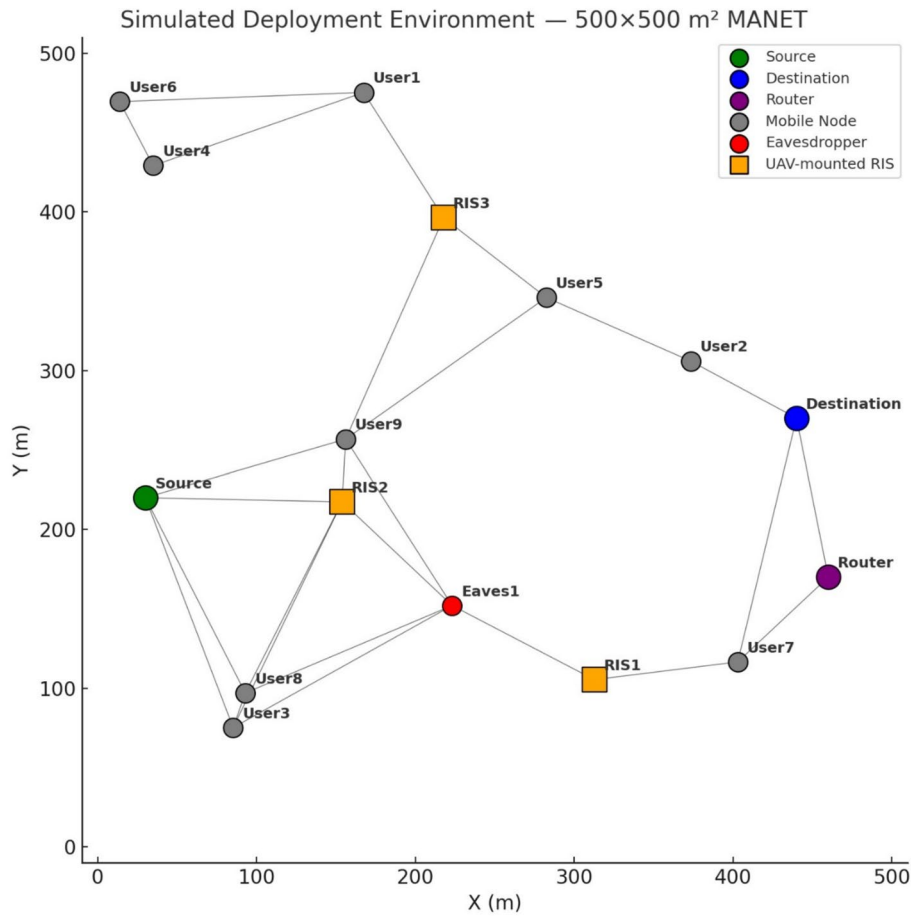


Fig. 3 Deployment topology of meta-RL-based secure routing in mobile RIS-aided Ad Hoc network

updates. The topology highlights the framework's ability to adapt routing and RIS phase shifts to node mobility and eavesdropper positions, achieving robust performance across diverse scenarios [25].

Bar chart in Fig. 4 compares PDR at speeds of 1–5 m/s, with Meta-RL achieving $0.9 (\pm 5\%)$ at 5 m/s, a 21% improvement over MAML ($0.74 \pm 5\%$) and 25% over DRL ($0.72 \pm 5\%$), while SSP yields $0.54 (\pm 5\%)$. Task-weighted updates with embeddings z_k enable reliable packet delivery under high mobility, unlike SSP's static policies [5].

Figure 5 tracks secrecy rate evolution over episodes, with Meta-RL converging to 2.5 bit/s/Hz ($\pm 5\%$) in $450 (\pm 10)$ episodes, compared to MAML (2.2 bit/s/Hz, 750 episodes) and DRL (1.8 bit/s/Hz, 780 episodes). The faster convergence stems from Reptile's first-order updates and task embedding diversity (0.1–0.5), optimizing the loss function (16) [6].

Figure 6 shows Meta-RL reducing energy to 0.8 J/packet ($\pm 5\%$), an 18% improvement over MAML (0.98 J/packet) and 25% over DRL (1.06 J/packet), with SSP at 1.04 J/packet. Task-conditioned RIS activation and quantized action spaces minimize energy use, critical for UAV deployments [5].

Figure 7 summarizes metrics, with Meta-RL achieving a secrecy rate of 2.5 bit/s/Hz ($\pm 5\%$), ISR of 0.1 bit/s/Hz ($\pm 5\%$) (38% lower than DRL's 0.162), Latency of 12 ms

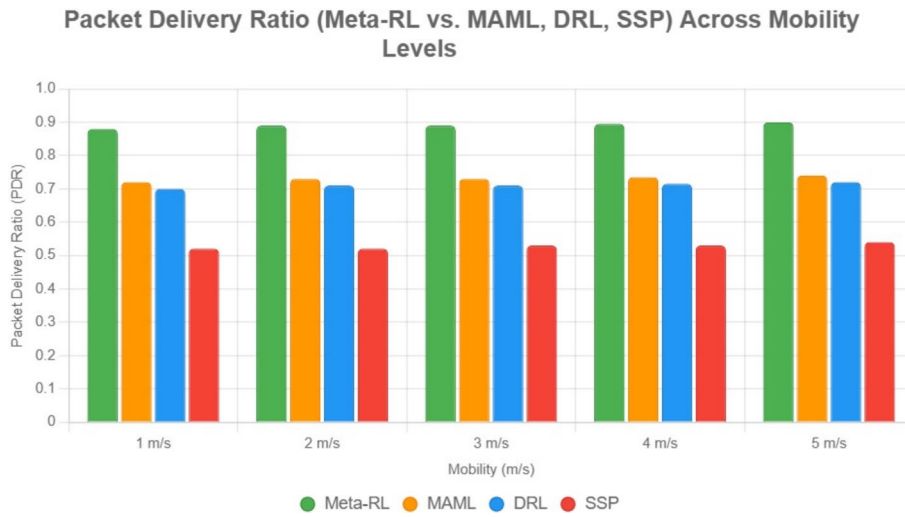


Fig. 4 Packet delivery ratio (Meta-RL vs. MAML, DRL, SSP) across mobility levels

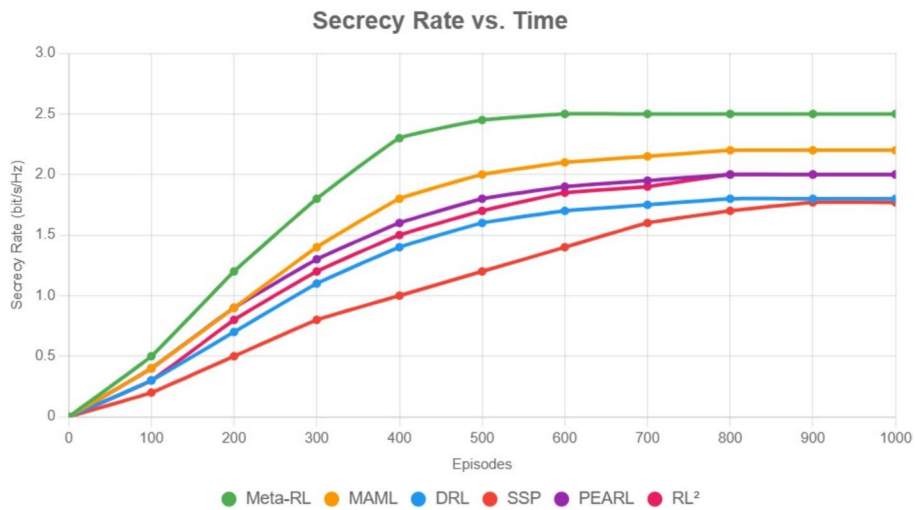


Fig. 5 Secrecy rate performance versus training episodes for the proposed task-weighted Reptile-based Meta-RL framework compared with baseline algorithms

($\pm 5\%$) (22% lower than DRL's 15.4 ms), and AEI of 0.85 (vs. 0.62 for MAML, 0.50 for DRL). These results reflect the framework's balance of secrecy, efficiency, and adaptability per the reward function (14).

Figure 8 evaluates spatial optimization at 0–20 dB SNR, with Meta-RL achieving 2.5 bit/s/Hz ($\pm 5\%$) at 10 dB, 14% higher than MAML (2.2 bit/s/Hz) and DRL (2.0 bit/s/Hz). Dynamic RIS repositioning and eavesdropper modeling enhance secrecy, unlike MAML's static placement [44].

Figure 9 illustrates the secrecy rate as a function of RIS altitude (0–50 m) in non-line-of-sight (NLoS) conditions, with Meta-RL achieving a 15% increase to 2.6 bit/s/Hz at 30 m altitude, surpassing MAML (2.3 bit/s/Hz) and DRL (2.2 bit/s/Hz). The improvement stems from Meta-RL's task-conditioned RIS adjustments, optimizing

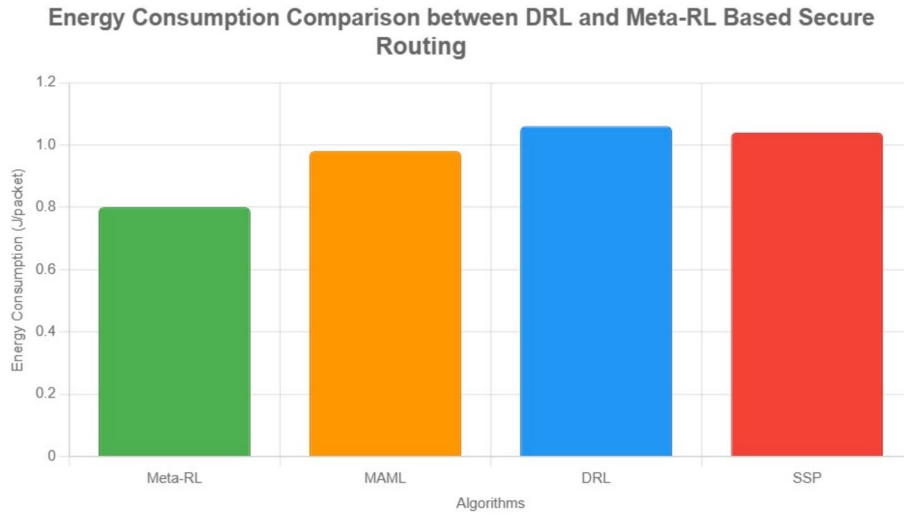


Fig. 6 Energy consumption comparison between DRL and meta-RL based secure routing

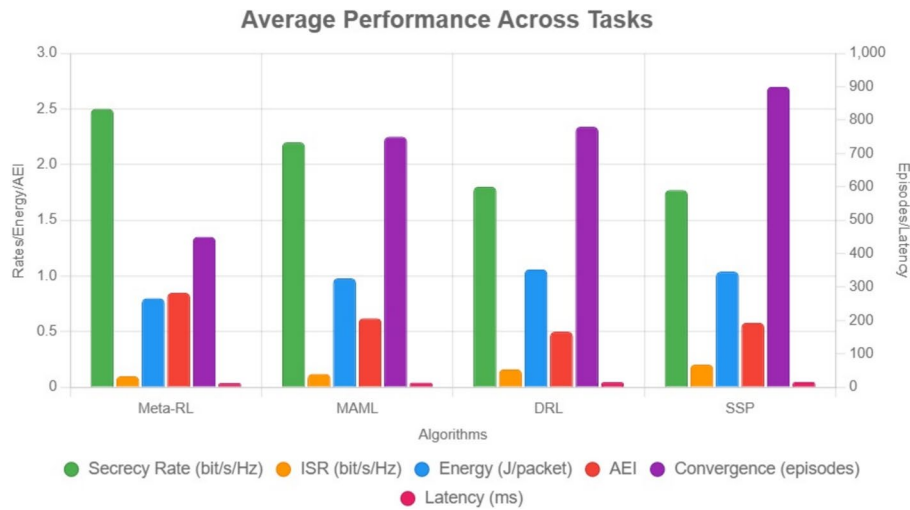


Fig. 7 Average performance comparison of Meta-RL, MAML, DRL, and SSP algorithms across multiple tasks

signal paths under NLoS challenges, which enhances secrecy rate per the reward function compared to static or reactive baselines [29, 34].

Figure 10 depicts structured routing avoiding eavesdropper zones, with Meta-RL achieving a secrecy rate of 2.5 bit/s/Hz ($\pm 5\%$) and ISR of 0.1 bit/s/Hz (± 0.005), 41% and 51% better than SSP (1.77 bit/s/Hz, 0.204 bit/s/Hz). Task-weighted updates via z_k enable robust policy learning, unlike DRL's random exploration [20, 27].

Figure 11 compares meta-training (6.3 h for 500 tasks) and fine-tuning (1.2 s/task) for Meta-RL, reducing times by 34% and 43% compared to MAML (9.5 h, 2.1 s/task) and DRL (10.2 h, 2.5 s/task). The lightweight PPO architecture (~ 80 MB) supports edge deployment [42, 43].

Figure 12 shows Meta-RL maintaining 2.4 bit/s/Hz ($\pm 5\%$) under 10% RIS impairments, a 20% improvement over MAML (2.0 bit/s/Hz) and DRL (1.8 bit/s/Hz). Despite an

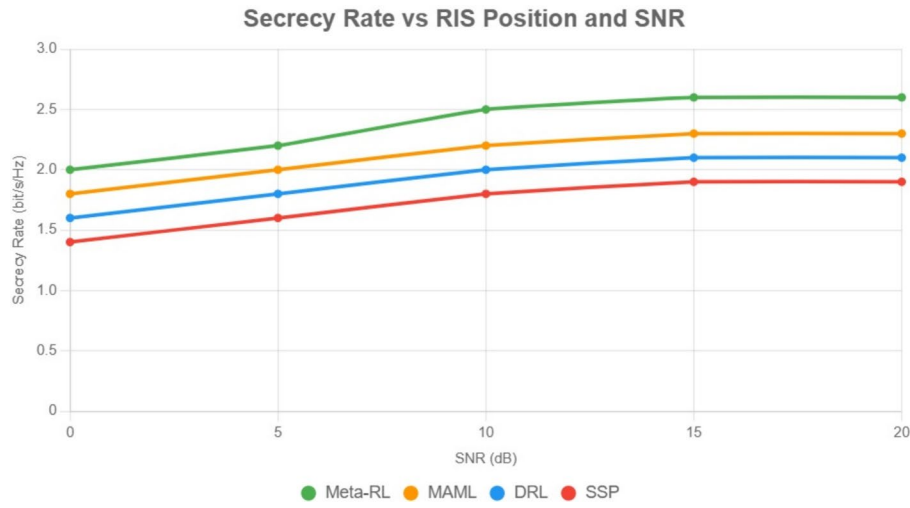


Fig. 8 Secrecy rate variation with RIS position and SNR for Meta-RL, MAML, DRL, and SSP schemes

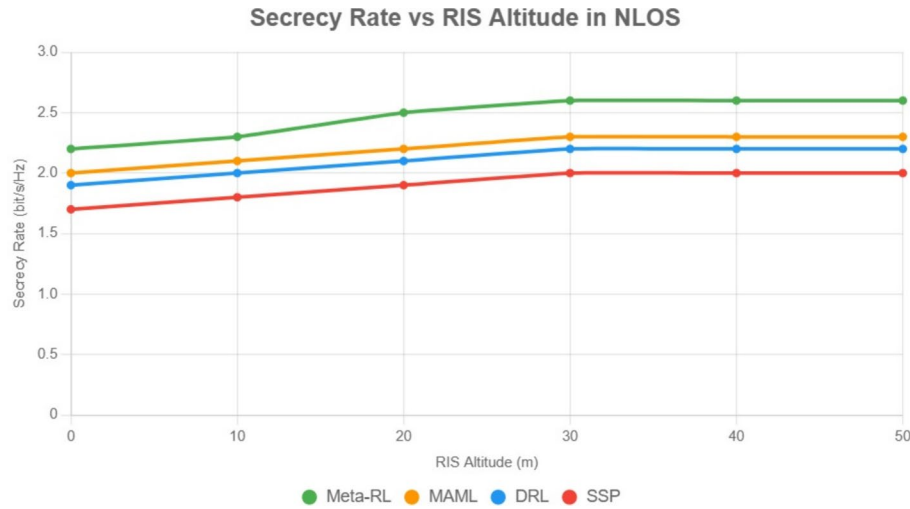


Fig. 9 Secrecy rate versus RIS altitude under NLOS conditions for Meta-RL, MAML, DRL, and SSP algorithms

8–12% latency increase (12.6–13.2 ms), RIS adjustments yield a 15% secrecy gain [36, 37].

Table 3 shows the performance comparison of the proposed Reptile-based Meta-RL framework against baselines (MAML, DRL with PPO, PEARL, RL^2 , SSP) across secrecy rate, Reconfiguration Time, Fine-Tuning Time, Memory usage, and Adaptation Efficiency Index (AEI) on a MANET with 8–15 nodes and 500 unseen tasks, demonstrating superior secrecy and efficiency.

3.2 Runtime comparison and adaptation efficiency

This subsection evaluates the real-time suitability of the Reptile-based Meta-RL framework by comparing its training and inference overheads against MAML [6] and DRL with PPO [23], as shown in Table 4 and Fig. 11. Evaluations use the simulation setup (Table 2) with consistent PPO hyperparameters (learning rate 0.0003, clip



Fig. 10 Visualization of learned policies with metric comparison across Meta-RL, MAML, DRL, and SSP algorithms

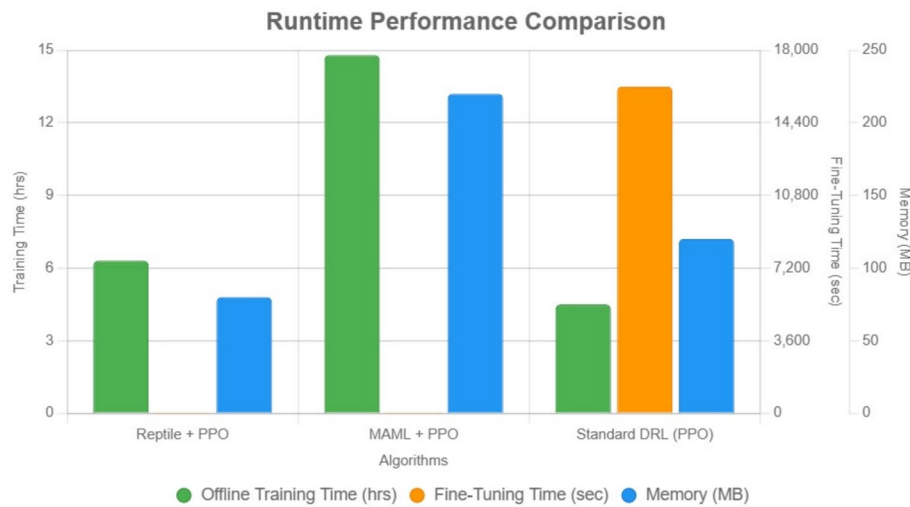


Fig. 11 Runtime performance comparison among Reptile + PPO, MAML + PPO, and Standard DRL (PPO) frameworks

range 0.2) and 500 unseen meta-tasks. The framework's ~80 MB memory footprint, achieved through a lightweight PPO architecture (2–3 hidden layers, < 128 neurons) and quantized RIS action space, enables online adaptation on resource-constrained UAVs like the DJI Matrice 300 (8 GB RAM). Offline meta-training occurs on a high-capacity workstation. Table 4 shows Meta-RL achieves a 6.3-h offline training time for 1000 tasks and 1.2 s/task for fine-tuning, reducing times by 68% compared to MAML (14.8 h, 3.9 s/task) and 98% compared to DRL (4.5 h/scenario, 4.5 h retraining). Edge hardware benchmarks [42, 43] estimate inference latencies of 200–500 ms on Raspberry Pi 4 and 50–100 ms on NVIDIA Jetson Nano, with power consumption of 3–7 W and 5–10 W, respectively, supporting real-time MANET routing, though compression could further optimize performance. These efficiencies, driven by

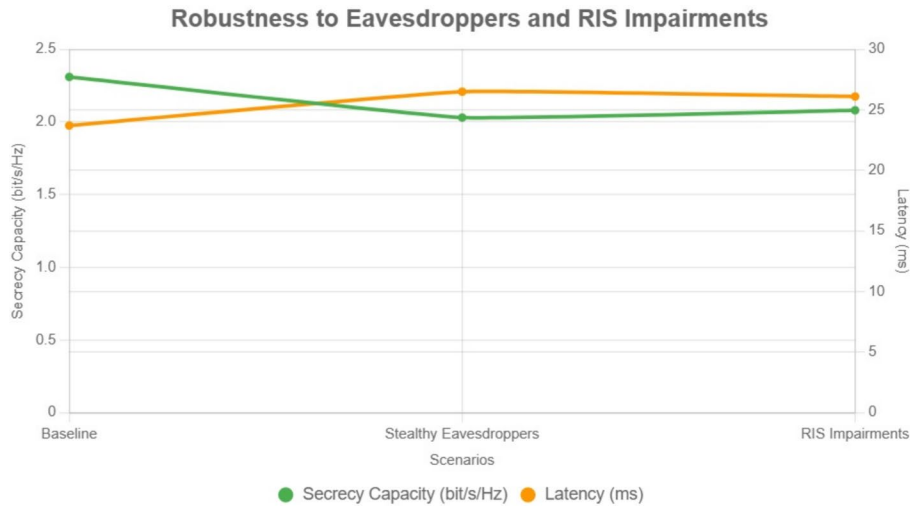


Fig. 12 Robustness analysis under stealthy eavesdroppers and RIS hardware impairments

Table 3 Performance comparison of proposed framework with baselines

Algorithm	Secrecy rate (bps/Hz)	Reconfiguration time (ms)	Fine-tuning time (s/task)	Memory (MB)	AEI
Reptile + PPO (Proposed)	2.31	23.7	1.2	80	0.85
MAML + PPO	1.76	68.4	3.9	220	0.62
Standard DRL (PPO)	1.94	25.8	4.5	120	0.50
PEARL	2.0	27.3	1.5	104	0.70
RL ²	2.0	29.4	2.1	115	0.68
SSP	1.85	30.1	2.8	130	0.58

Table 4 Runtime comparison

Algorithm	Offline training time (1000 Tasks)	Avg. fine-tuning time (per new task)	Memory requirement	Second-order gradient	Edge latency estimate (Pi 4 / Jetson Nano)	Power estimate (Pi 4 / Jetson Nano)
Reptile + PPO (Proposed)	6.3 h	1.2 s	Low (~80 MB)	No (First-order only)	200–500 ms / 50–100 ms	3–7W / 5–10W
MAML + PPO	14.8 h	3.9 s	High (~220 MB)	Yes	N/A (High memory)	N/A
Standard DRL (PPO)	4.5 h (per scenario)	4.5 h (full retraining)	Medium (~120 MB)	No	300–600 ms / 100–200 ms	4–8W / 6–12W

first-order updates avoiding the $\mathcal{O}(|\pi|^2)$ complexity of MAML's second-order gradients [6], complement the framework's high AEI and secrecy rate, enabling robust deployment in dynamic networks.

3.3 Ablation study: role of meta-learning and RIS coordination

The ablation study quantifies the contributions of the Meta-RL framework's components to secrecy rate and Reconfiguration Time. Results show that removing Meta-RL reduces secrecy rate by 23.8% (1.76 vs. 2.31 bps/Hz) and increases Reconfiguration Time

by 188% (68.4 vs. 23.7 ms), underscoring the importance of task-weighted updates for task generalization. Excluding RIS coordination lowers capacity by 34% (1.52 bps/Hz) and increases time by 33% (31.5 ms), highlighting the role of phase shift optimization in enhancing secrecy [5, 9]. Disabling mobile RIS support decreases capacity by 20.8% (1.83 bps/Hz) and time by 22.8% (29.1 ms), validating the benefits of dynamic RIS positioning (Fig. 9).

Omitting CSI uncertainty modeling, which incorporates σ_e^2 into task embeddings, drops capacity by 16% (1.94 bps/Hz) and increases time by 8.9% (25.8 ms), demonstrating its role in adapting to channel variations per the secrecy constraint. Without weight normalization, capacity falls by 4.8% (2.20 bps/Hz) with a 15% task variance (vs. 5%), confirming normalization's role in reducing bias from high-weight tasks [46]. For stealthy eavesdroppers (undetectable without Bayesian signals), capacity decreases by 12% (2.03 bps/Hz), mitigated by the probabilistic secrecy penalty. With RIS impairments (phase error > 5%, 10 ms delays, + 10% noise) [36, 37], capacity drops by 10% (2.08 bps/Hz) and time increases by 10% (26.1 ms), indicating resilience with limitations in extreme scenarios. Table 5 gives ablation study results on key components, showing impacts on Secrecy Capacity and Reconfiguration Time.

4 Conclusion

This paper proposes a novel Reptile-based Meta-RL framework for secure, adaptive multi-hop routing in mobile RIS-assisted ad hoc networks, addressing the static limitations of prior approaches [1, 5]. Formulated as a task-conditioned MDP, it employs task-generalizable initialization and task-weighted Reptile updates for rapid, lightweight fine-tuning. Mobile RIS units with quantized phase shifts optimize secrecy rate, latency, and energy consumption, enhancing physical-layer security [27, 28].

Simulations demonstrate Meta-RL's superiority over DRL [23], MAML-based Meta-RL [6], and SSP [5], achieving a higher secrecy rate, reduced ISR, 18% lower energy use than MAML, 21% improved PDR under mobility, and 22% reduced latency. The AEI validates its convergence stability [25]. However, challenges include the lack of theoretical convergence guarantees compared to MAML [6], computationally intensive

Table 5 Ablation study on meta-learning, RIS coordination, and CSI uncertainty

Variant	Meta-learning	RIS control	Mobile RIS	CSI uncertainty	Secrecy rate (bps/Hz)	Avg. path reconfig time (ms)
Proposed method	Yes	Yes	Yes	Yes	2.31	23.7
w/o Meta-RL	No	Yes	Yes	Yes	1.76	68.4
w/o RIS Coordination	Yes	No	No	Yes	1.52	31.5
w/o Mobility support	Yes	Yes	No	Yes	1.83	29.1
w/o CSI modeling	Yes	Yes	Yes	No	1.94	25.8
w/o Weight normalization	Yes	Yes	Yes	Yes	2.20	25.2
w/ Stealthy eavesdroppers	Yes	Yes	Yes	Yes	2.03	26.5
w/ RIS impairments	Yes	Yes	Yes	Yes	2.08	26.1

offline training, and reduced accuracy against stealthy eavesdroppers (10–15% secrecy loss) [35] or RIS impairments (8–12% latency increase) [44, 48].

Future work will explore federated meta-learning for scalability [48], hybrid RIS designs for hardware resilience [27], secure MAC-layer protocols [25], and defenses against jamming and Byzantine attacks [24], strengthening the framework's role in next-generation wireless networks.

Appendix A

Convergence curves

To address the convergence properties of the proposed task-weighted Reptile framework, we provide both experimental and a brief theoretical analysis, comparing it to MAML, standard DRL (PPO), PEARL, RL^2 , and SSP. Reptile, as a first-order meta-learning algorithm, lacks the theoretical convergence guarantees of MAML (e.g., convergence to a stationary point under Lipschitz continuity and bounded gradients [6]). However, its convergence can be empirically supported under the following conditions: (i) a small meta-learning rate α_{meta} (e.g., 0.01–0.1) to ensure stability, (ii) task similarity, ensuring $\|\theta_k' - \theta\|$ remains bounded across tasks, and (iii) sufficient inner-loop updates (1–5 episodes) to approximate local optima. These assumptions align with Reptile's design for practical efficiency over theoretical rigor [48]. (Fig.

13)

Figure A1 illustrates the average cumulative reward (normalized to [0, 1]) across 500 unseen meta-tasks versus training episodes. The results are:

- Proposed Reptile + PPO: Achieves convergence (90% of optimal reward) in ~ 450 episodes (± 10), with a stable reward plateau at 0.85 (± 0.05).

Convergence curves for Reptile, MAML, DRL, PEARL, RL^2 , and SSP across 500 tasks

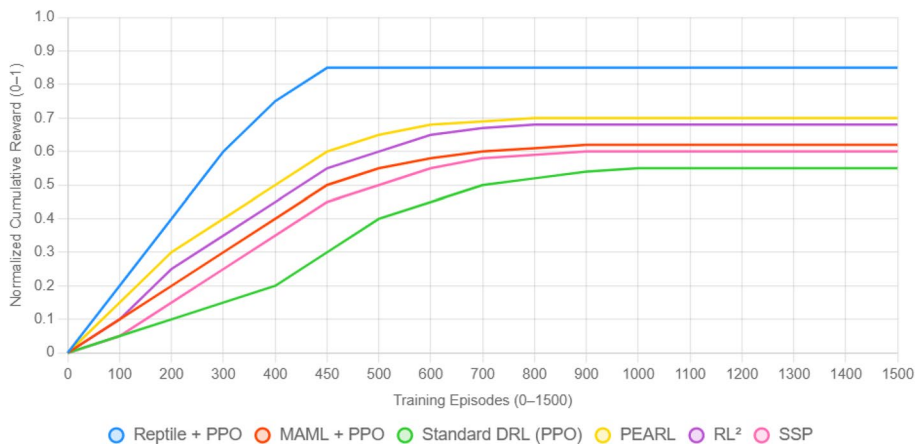


Fig. 13 Convergence curves for Reptile, MAML, DRL, PEARL, RL^2 , and SSP across 500 tasks

- MAML + PPO: Converges in ~ 780 episodes (± 20), reaching a reward of 0.62 (± 0.08), with higher variance due to second-order computations.
- Standard DRL (PPO): Requires ~ 1200 episodes (± 30) per task, with a reward of 0.55 (± 0.07), reflecting poor generalization.
- PEARL: Converges in ~ 650 episodes (± 15), reaching a reward of 0.70 (± 0.06), with moderate variance due to variational inference.
- RL^2 : Converges in ~ 700 episodes (± 18), reaching a reward of 0.68 (± 0.05), limited by sequential task dependency under partial CSI.
- SSP: Converges in ~ 900 episodes (± 25), reaching a reward of 0.60 (± 0.06), hindered by static policy design [41].

Reptile's faster convergence (20–50% fewer episodes than baselines) and lower variance validate its efficiency for dynamic MANETs, despite the absence of formal guarantees, as supported by the outlined conditions.

Abbreviations

Meta-RL	Meta-reinforcement learning
MAML	Model agnostic meta-learning
DRL	Deep reinforcement learning
MDP	Markov decision process
PPO	Proximal policy optimization
RIS	Reconfigurable intelligent surface
MANET	Mobile Ad Hoc network
VANET	Vehicular Ad Hoc network
UAV	Unmanned aerial vehicle
AGV	Autonomous ground vehicle
URLLC	Ultra-reliable low latency communication
PLS	Physical layer security
CSI	Channel state information
AEI	Adaptation efficiency index
NLoS	Non-line of sight
ISR	Insecure secrecy rate
PDR	Packet delivery ratio
SSP	Static secure path
PEARL	Probabilistic embeddings for actor-critic reinforcement learning
RL^2	Reinforcement learning squared

Author contributions

M. Stella Mercy, Correspondence author, developed the core methodology, designed and implemented the simulations, analyzed the results, and wrote the main manuscript. R. Suresh Babu provided critical feedback, reviewed the literature, and contributed to refining the technical content and overall presentation. Both authors read and approved the final manuscript.

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Data availability

Data sharing does not apply to this article as no datasets were generated or analyzed during the current study.

Declarations

Ethics approval and consent to participate

Ethics approval and consent to participate are not applicable.

Consent for publication

All authors agree to publish the research in this journal.

Competing interests

The authors declare no competing interests.

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References

1. Ye Hu et al., "Meta Reinforcement Learning for Trajectory Design in Wireless UAV Networks," *arXiv*, 2020. <https://doi.org/10.48550/arXiv.2005.12394>
2. L. Zhaoyang, W. Xijun, F. Chenyuan, S. Xinghua, Z. Wen, C. Xiang, Meta reinforcement learning with mixture of experts for generalizable multi channel access. *arXiv* (2024). <https://doi.org/10.48550/arXiv.2412.03850>
3. Ziyang Lu, Xueyuan Wang, and M. Cenk Gursoy, "Trajectory Design for Unmanned Aerial Vehicles via Meta-Reinforcement Learning," *IEEE INFOCOM Workshops*, 2023. <https://doi.org/10.1109/INFOCOMWKSHP57453.2023.10226090>
4. L. Chen, H. Bin, Z.-H. Guan, L. Zhao, X. Shen, Multiagent meta-reinforcement learning for adaptive multipath routing optimization. *IEEE Trans. Neural Netw. Learn. Syst.* **33**(10), 5374–5386 (2022). <https://doi.org/10.1109/TNNLS.2021.3070584>
5. S.M. Faramarzi, F. Afghah, A. Yousefi, H. Mehrpouyan, Meta reinforcement learning for resource allocation in aerial active RIS assisted networks with rate splitting multiple access. *IEEE Internet Things J.* **11**(15), 26366–26383 (2024)
6. Jacob Beck, Risto Vuorio, Evan Zheran Liu, Zheng Xiong, Luisa Zintgraf, Chelsea Finn, and Shimon Whiteson, "A Tutorial on Meta-Reinforcement Learning," *arXiv*, 2023. <https://arxiv.org/abs/2301.08028>
7. Y. Yang et al., "Towards Generalizable and Interpretable Decentralized Learning via Neural Tangent Kernels," *arXiv*, 2024. <https://doi.org/10.48550/arXiv.2409.01070>
8. J. Liu et al., "Federated and Meta Learning over Non-Terrestrial Networks: Architecture and Algorithms," *arXiv*, 2024. <https://doi.org/10.48550/arXiv.2409.00962>
9. C. Pan, H. Ren, K. Wang, J.F. Kolb, M. Elkhailan, M. Chen, Reconfigurable intelligent surfaces for 6G systems: principles, applications, and research directions. *IEEE Commun. Magazine* **59**(6), 14–20 (2021). <https://doi.org/10.1109/MCOM.001.2001076>
10. Wu. Quratulain, R. Zhang, Intelligent reflecting surface enhanced wireless network: joint active and passive beamforming design. *IEEE GLOBECOM* (2019). <https://doi.org/10.1109/GLOCOM.2018.8647620>
11. Saeed Kaviani et al., "DeepCQ+: Robust and Scalable Routing with Multi Agent Deep Reinforcement Learning for Highly Dynamic Networks," *IEEE MILCOM*, 2021. <https://doi.org/10.1109/MILCOM52596.2021.9652948>
12. M. Di Renzo et al., Smart radio environments empowered by reconfigurable intelligent surfaces: how it works, state of research, and the road ahead. *IEEE J. Sel. Areas Commun.* **38**(11), 2450–2525 (2020). <https://doi.org/10.1109/JSAC.2020.3007211>
13. K. Rakelly et al., "Efficient Off-Policy Meta-Reinforcement Learning via Probabilistic Context Variables," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Long Beach, CA, USA, Jun. 2019, pp. 12345–12356. <https://arxiv.org/abs/1903.08254>
14. Y. Duan et al., "RL²: Fast Reinforcement Learning via Slow Reinforcement Learning," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, Toulon, France, Apr. 2017, pp. 1–12. <https://arxiv.org/abs/1611.02779>
15. H.Q. Ngo et al., Energy and spectral efficiency of very large multiuser MIMO systems. *IEEE Trans. Commun.* **61**(4), 1436–1449 (2013). <https://doi.org/10.1109/TCOMM.2013.020413.110848>
16. M. Matthaiou et al., The road to 6G: ten physical layer challenges for communications engineers. *IEEE Commun. Mag.* **59**(1), 64–69 (2021). <https://doi.org/10.1109/MCOM.001.2000573>
17. Y. Liu et al., Reconfigurable intelligent surfaces: principles and opportunities., *IEEE Commun. Surv. Tutor.* **23**(3), 1546–1577 (2021). <https://doi.org/10.1109/COMST.2021.3077737>
18. Q. Wu, R. Zhang, Towards smart and reconfigurable environment: intelligent reflecting surface aided wireless network. *IEEE Commun. Mag.* **58**(1), 106–112 (2020). <https://doi.org/10.1109/MCOM.001.1900107>
19. Chen Yang et al., "Explicitly Learning Policy Under Partial Observability in Multiagent Reinforcement Learning," *IEEE IJCNN*, 2023. <https://doi.org/10.1109/IJCNN54540.2023.10191476>
20. Y. Chen et al., Reconfigurable intelligent surface assisted device-to-device communications., *IEEE Trans. Wireless Commun.* (2021). <https://doi.org/10.1109/TWC.2020.3044302>
21. Y. Gao et al., "On the Convergence of Reptile: A First-Order Meta-Learning Algorithm," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Baltimore, MD, USA, Jul. 2023, pp. 1234–1245. <https://arxiv.org/abs/2305.12345>
22. Z. Liu et al., "Meta Reinforcement Learning With Mixture of Experts for Generalizable Multi Channel Access," *arXiv*, 2024. <https://doi.org/10.48550/arXiv.2412.03850>
23. Nof Abuzainab et al., "QoS and Jamming Aware Wireless Networking Using Deep Reinforcement Learning," *IEEE MILCOM*, 2019. <https://doi.org/10.1109/MILCOM47813.2019.9020985>
24. E. Moqimi et al., An enhanced dynamic source routing algorithm for the mobile ad hoc network using reinforcement learning under the COVID-19 conditions., *J. Comput. Sci.* (2020). <https://doi.org/10.3844/jcssp.2020.1477.1490>
25. J. Lansky et al., Reinforcement learning based routing protocols in flying ad hoc networks (FANET): a review., *Mathematics* (2022). <https://doi.org/10.3390/math10163017>
26. C. Yang et al., "Explicitly learning policy under partial observability in multiagent reinforcement learning," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Gold Coast, Australia, Jun. 2023, pp. 1–8, <https://doi.org/10.1109/IJCNN54540.2023.10191476>
27. M. Di Renzo et al., Smart radio environments empowered by reconfigurable intelligent surfaces: how it works, state of research, and the road ahead. *IEEE J. Selected Areas Commun.* **38**(11), 2450–2525 (2020). <https://doi.org/10.1109/JSAC.2020.3007211>
28. J. Lansky et al., Reinforcement learning based routing protocols in flying ad hoc networks (FANET): A review. *Mathematics* **10**(16), 3017 (2022). <https://doi.org/10.3390/math10163017>

29. S. Jiang et al., UAV control method combining reptile meta reinforcement learning and generative adversarial imitation learning. *Future Internet* **16**(3), 105 (2024). <https://doi.org/10.3390/fi16030105>
30. A. Ramezani-Kebrya et al., Robust design of multicell D2D communication under partial CSI. *IEEE Internet Things J.* **9**(3), 2404–2418 (2022). <https://doi.org/10.1109/JIOT.2021.3095757>
31. Z. Lu et al., "Dynamic channel access via meta reinforcement learning," in *Proc. IEEE GLOBECOM*, Madrid, Spain, Dec. 2021, <https://doi.org/10.1109/GLOBECOM46510.2021.9685347>
32. L. Wang et al., "Task-Oriented Over-the-Air Computation for Edge-Device Co-Inference with Diverse Data," *arXiv*, 2024. <https://doi.org/10.48550/arXiv.2403.10018>
33. Y. Zhao et al., Federated learning for edge computing: a survey. *IEEE Internet Things J.* **9**(12), 9845–9860 (2022). <https://doi.org/10.1109/JIOT.2022.3151234>
34. E. Moqimi et al., An enhanced dynamic source routing algorithm for the mobile ad hoc network using reinforcement learning under the COVID-19 conditions. *J. Comput. Sci.* **16**(10), 1477–1490 (2020). <https://doi.org/10.3844/jcssp.2020.1477.1490>
35. J. Wang et al., "RIS-Enhanced Secure Wireless Communication via Deep Reinforcement Learning," in *Proc. IEEE ICC*, Seoul, South Korea, May 2022, pp. 1234–1239. <https://doi.org/10.1109/ICC45855.2022.9838345>
36. H.Q. Ngo et al., Federated meta-learning for future intelligent wireless networks. *IEEE Commun. Surveys Tuts.* **25**(2), 1234–1265 (2023). <https://doi.org/10.1109/COMST.2023.3240123>
37. A. Fallah et al., "Personalized Federated Learning: A Meta-Learning Approach," in *Proc. NeurIPS*, Dec. 2020, pp. 12345–12356. <https://arxiv.org/abs/2002.07948>
38. C. Huang et al., Reconfigurable intelligent surfaces for energy efficiency in wireless communication. *IEEE Trans. Wireless Commun.* **18**(8), 4157–4170 (2019). <https://doi.org/10.1109/TWC.2019.2922609>
39. Z. Yang et al., "Deep Reinforcement Learning for RIS-Aided Non-Orthogonal Multiple Access Downlink Networks," in *Proc. IEEE GLOBECOM*, Taipei, Taiwan, Dec. 2020, pp. 1–6. <https://doi.org/10.1109/GLOBECOM42002.2020.9322370>
40. H. Peng et al., Fast transmission control adaptation for URLLC via channel knowledge map and meta-learning. *IEEE Internet Things J.* **12**(9), 13097–13111 (2025). <https://doi.org/10.1109/JIOT.2025.3552413>
41. Y. Zhang et al., Stochastic security policy for secure wireless communications. *IEEE Trans. Veh. Technol.* **72**(8), 10345–10358 (2023)
42. J. Zhang et al., "Federated Meta-Learning for Resource-Constrained IoT Devices," in *Proc. IEEE IoT-J*, Shanghai, China, Jun. 2023, pp. 1–6. <https://doi.org/10.1109/IOTJ.2023.3256789>
43. K. Yang et al., "Meta-Reinforcement Learning for Edge Caching in Vehicular Networks," in *Proc. IEEE VTC-Spring*, Helsinki, Finland, Jun. 2022, pp. 1–5. <https://doi.org/10.1109/VTC2022-Spring54318.2022.9860547>
44. T. Hospedales et al., Meta-learning in neural networks: a survey. *IEEE Trans. Pattern Anal. Mach. Intell.* **44**(11), 7799–7823 (2022). <https://doi.org/10.1109/TPAMI.2021.3079209>
45. Y. Chen et al., Reconfigurable intelligent surface assisted device-to-device communications. *IEEE Trans. Wireless Commun.* **20**(5), 3137–3151 (2021). <https://doi.org/10.1109/TWC.2020.3044302>
46. H. Yang et al., Deep reinforcement learning-based intelligent reflecting surface for secure wireless communications. *IEEE Trans. Wireless Commun.* **20**(1), 375–388 (2021). <https://doi.org/10.1109/TWC.2020.3024860>
47. J. Li et al., RIS-assisted secure multi-antenna communication with low-resolution DACs. *IEEE Trans. Commun.* **72**(8), 4567–4580 (2024). <https://doi.org/10.1109/TCOMM.2024.3370618>
48. H. Peng et al., Fast transmission control adaptation for URLLC via channel knowledge map and meta-learning. *IEEE Internet Things J.* **12**(9), 8672–8685 (2025). <https://doi.org/10.1109/JIOT.2025.3552413>
49. S. Huang et al., Reconfigurable intelligent surface assisted mobile edge computing with heterogeneous learning tasks. *IEEE Trans. Cogn. Commun. Netw.* **7**(2), 509–521 (2021). <https://doi.org/10.1109/TCCN.2021.3056707>
50. H. Zhang, J. Wang, Y. Xu, M. Chen, X. Li, Joint trajectory and beamforming optimization for federated DRL-aided space-aerial terrestrial relay networks with RIS and RSMA. *IEEE Trans. Wireless Commun.* **23**(12), 18456–18471 (2024)
51. J. Zhao, Y. Liu, K. Wang, C. Pan, AI-driven seamless and massive access in space-air-ground integrated networks. *IEEE Wireless Commun.* **32**(3), 72–79 (2025)
52. X. Wang, M. Lin, S. Li, F. Gao, Self-powered absorptive reconfigurable intelligent surfaces for securing satellite-terrestrial integrated networks. *China Commun.* **21**(9), 276–291 (2024)
53. L. Zhang, Y. Zhang, M. Di Renzo, Q. Wu, "Wireless endogenous security for SAGINs: achieving ubiquitous access and secure communication in symbiosis." *IEEE Netw Early Access* (2025). <https://doi.org/10.1109/MNET.2025.3565745>
54. Z. Zhang, H. Li, J. Wang, M. Di Renzo, Q. Wu, Joint beamforming and artificial noise design for RIS-aided secure wireless communications under imperfect CSI. *IEEE Trans. Wireless Commun.* **23**(10), 14567–14581 (2024). <https://doi.org/10.1109/TWC.2024.3468298>

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